



EAST WEST UNIVERSITY

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Description

Student Hostel Management System is a database-driven tool that supports efficient storage of student personal information, hostel details, hostel staff basic information , student payments, room and bed details and other compulsory information. Each hostel has a unique ID and multiple floors. Additionally, each floor has a unique floor ID with multiple rooms. Furthermore, each room has room ID with multiple or single beds. The rent for each bed is categorized based on its condition. One student can enter any hostel based on availability of bed. When one student gets into any hostel then allocation table would create. It contains a unique allocation id , start and end date of every student of their hostel journey. Whenever any problem arises about hostel student can complain. On the other hand, students can make maintenance requests. One staff member can manage multiple maintenance requests, and one maintenance request may need more than one staff member. This system records all data without any hassle also this is very much user-friendly system. By using the database management system any hostel can manage all their functions in more dynamic way.

Entity And Attributes

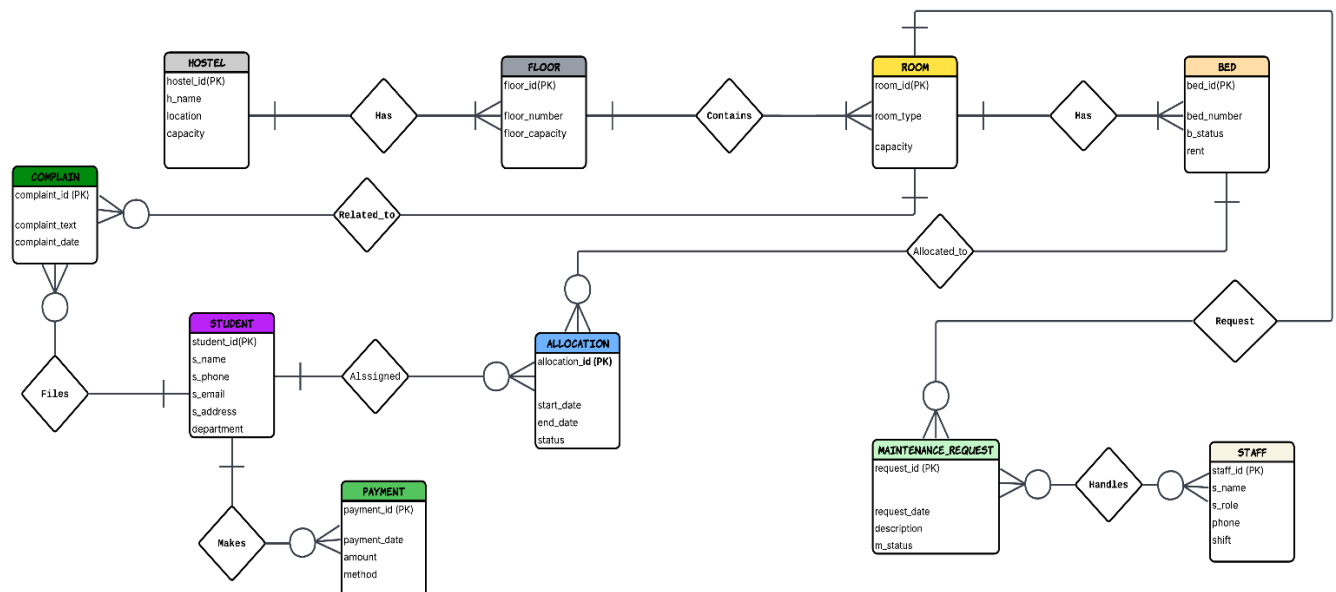
Entity	Attributes
Hostel	hostel_id(pk), hostel_name, location, capacity
Floor	floor_id(pk), floor_number, floor_capacity
Room	room_id(pk), room_type, room_capacity
Bed (week)	bed_id(pk), bed_number, b_status, rent
Student	student_id(pk), s_name, s_phone ,s_email ,s_address ,department
Complain	complaint_id(pk), complaint_text ,complaint_date
Payment	payment_id(pk), payment_date ,amount ,method
Allocation	allocation_id(pk), start_date, end_date, status
Staff	staff_id(pk), staff_name, staff_role, phone
Maintenance_Request	request_id(pk), request_date, description, m_status

Relationship

Relation	Connected Entity	Cardinality	Meaning
Has	Hostel - Floor	1:N(Mandatory)	One hostel must have one or more floors, and each floor belongs to one hostel.
Contains	Floor - Room	1:N(Mandatory)	One floor must contain one or more rooms, and each room belongs to one floor.
Has	Room - Bed	1:N(Mandatory)	One room can have one or many beds and every bed belongs to one room.
Makes	Student - Payments	1:N(Optional)	One student can payment zero, one or many and each payment must refer to one student.
Request	Room - Maintenance_Request	1:N(Optional)	one room may request zero or many maintenance requests and each request must contain a room.
Handles	Staff - Maintenance_Request	M:N(Optional)	One staff member may involve in zero or many maintenances and each maintenance may contain one or many staff.
Files	Student - Complain	1:N(Optional)	One student may file zero ,one or many complaints and every complaint belongs to one student.
Related_To	Room - Complain	1:N(Optional)	One room can have zero or more complaints, and

			each complaint must refer a room.
Assigned	Student - Allocation	1:N(Optional)	One student may have zero or many allocations over time and each allocation must belong to one student
Allocated_to	Bed - Allocation	1:N(Optional)	One bed can participate in many allocations, and every allocation must refer to a bed.

Entity Relationship Diagram (ERD)



Relational Schema

- 1) HOSTEL** (hostel_id(PK), h_name, capacity, location)
- 2) FLOOR** (floor_id(PK), hostel_id(FK), floor_number, floor_capacity)
- 3) ROOM** (room_id(PK), floor_id(FK), room_type, room_capacity)
- 4) BED** (bed_id(PK), room_id(FK), bed_number, bed_status, rent)
- 5) STUDENT** (student_id(PK), s_name, s_email, s_phone, s_address, department)
- 6) COMPLAIN** (complaint_id(PK), student_id(FK), room_id(FK), complaint_text, complaint_date)
- 7) PAYMENT** (payment_id(PK), student_id(FK), payment_date, amount, method)
- 8) ALLOCATION** (allocation_id(PK), student_id(FK), bed_id(FK), start_date, end_date, status)
- 9) STAFF** (staff_id(PK), staff_name, staff_role, phone, shift)
- 10) MAINTENANCE_REQUEST** (request_id(PK), room_id(FK), request_date, description, m_status)
- 11) STAFF_MAINTENANCE** (staff_id(FK), request_id(FK))

Normalization

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. There are multiple phases in normalization like 1NF, 2NF, 3NF, BCNF, 4NF, 5NF.

1NF(First Normal Form):

Rule: Each column must contain **atomic** (indivisible) values, no repeating groups or arrays.

Analysis: All attributes in every column are atomic no multivariate attributes exists here.

Decision: All relationships satisfied 1NF

2NF (Second Normal Form):

Rule: All relations must be in 1NF. All non-key attributes must be fully functionally dependent on the entire primary key(No partial dependency).

Analysis: All tables use single -attribute primary keys (Not any composite key)

Decision: All relationships satisfied 2NF.

3NF(Third Normal Form):

Rule: Must be in 2NF. No transitive dependencies and all attributes must depend directly on the primary key

Analysis: All attributes directly depend on the primary key.

Decision: All tables' connections satisfy the 3NF.

BCNF(Boyce-Codd Normal Form):

Rule: Must be in 3NF. For every functional dependency $X \rightarrow Y$, X must be a super key and eliminates all anomalies due to functional dependencies.

Analysis: All tables have one primary key that determines all attributes.

Decision: This Relational Schema Satisfy BCNF.

4NF(Fourth Normal Form):

Rule: A relation is in Fourth Normal Form (4NF) if no multi-valued dependencies exist in relational schema.

Analysis: M:N relationships handled using separate tables ,

STAFF_MAINTENANCE (staff_id(FK), request_id(FK))

Decision: So, the relational schema satisfy 4NF.

The database relationships tables are normalized up to 4NF that ensures minimum data redundancy, improves data integrity, prevents update anomalies, and makes queries more efficient.

Implementation of Database

Structured Query Language (SQL) is the standard, universal language used for interaction with Database Management Systems (DBMS). It is the primary means by which all programs and users access data stored in a relational database. For this project we create a database called "student_hostel_management" by the command `CREATE DATABASE student_hostel_management;` and use DDL(Data Definition Language) and DML(Data Manipulation Language). DDL is mainly used to make a structure for a database. On the other hand, DML is used to fill the table that is made by DDL. We have 11 tables and some examples of creating tables by using DDL:

Run SQL query/queries on database student_hostel_management: ?

```
1 CREATE TABLE HOSTEL (  
2     hostel_id INT PRIMARY KEY,  
3     h_name VARCHAR(100),  
4     capacity INT,  
5     location VARCHAR(100)  
6 );  
7
```

Code for creating HOSTEL table

Run SQL query/queries on database student_hostel_management: ?

```
1 CREATE TABLE STUDENT (  
2     student_id INT PRIMARY KEY,  
3     s_name VARCHAR(100),  
4     s_email VARCHAR(100),  
5     s_phone VARCHAR(20),  
6     s_address VARCHAR(150),  
7     department VARCHAR(100)  
8 );  
9
```

Code for creating STUDENT table

Run SQL query/queries on database student_hostel_management: ?

```
1 CREATE TABLE PAYMENT (  
2     payment_id INT PRIMARY KEY,  
3     student_id INT,  
4     payment_date DATE,  
5     amount DECIMAL(10,2),  
6     method VARCHAR(50),  
7     FOREIGN KEY (student_id) REFERENCES STUDENT(student_id)  
8 );|
```

Code for creating PAYMENT table

Some examples of implementing data into student_hostel_management database:

Run SQL query/queries on database student_hostel_management: ?

```
1 INSERT INTO STUDENT VALUES  
2 (1,'Riyad','r1@mail.com','0171111','Dhaka','CSE'),  
3 (2,'Arafat','r2@mail.com','0172222','Gazipur','EEE'),  
4 (3,'Siam','r3@mail.com','0173333','Khulna','BBA'),  
5 (4,'Nafi','r4@mail.com','0174444','Dhaka','CSE'),  
6 (5,'Tanvir','r5@mail.com','0175555','Rajshahi','EEE'),  
7 (6,'Rakib','r6@mail.com','0176666','Sylhet','BBA'),  
8 (7,'Hasan','r7@mail.com','0177777','Barisal','CSE'),  
9 (8,'Fahim','r8@mail.com','0178888','Comilla','EEE'),  
10 (9,'Sakib','r9@mail.com','0179999','Dhaka','BBA'),  
11 (10,'Imran','r10@mail.com','0170000','Noakhali','CSE'),  
12 (11,'Jahid','j1@mail.com','0181112','Dhaka','CSE'),  
13 (12,'Rafi','j2@mail.com','0181113','Gazipur','EEE'),  
14 (13,'Mehedi','j3@mail.com','0181114','Khulna','BBA'),  
15 (14,'Shuvo','j4@mail.com','0181115','Sylhet','CSE'),
```

Data implementation on STUDENT table

Run SQL query/queries on database student_hostel_management: ?

```
1 INSERT INTO COMPLAINT VALUES
2 (1,1,101,'Fan not working','2026-02-10'),
3 (2,2,102,'Light problem','2026-02-12'),
4 (3,3,201,'Water problem','2026-02-14'),
5 (4,4,301,'Door lock broken','2026-02-15'),
6 (5,5,102,'Fan noise','2026-02-16');
7 |
```

Data implementation on COMPLAINT table

Run SQL query/queries on database student_hostel_management: ?

```
1 INSERT INTO STAFF VALUES
2 (1,'Rahim','Technician','0181111','Morning'),
3 (2,'Karim','Electrician','0182222','Evening'),
4 (3,'Salam','Plumber','0183333','Morning'),
5 (4,'Jamal','Cleaner','0184444','Night');|
```

Data implementation on STAFF table

Sample SQL Queries

Q1) Show all available rooms with hostel names

Run SQL query/queries on database `student_hostel_management`:

```
1 SELECT h.h_name, r.room_id, b.bed_id, b.bed_number
2 FROM HOSTEL h
3 JOIN FLOOR f ON h.hostel_id = f.hostel_id
4 JOIN ROOM r ON f.floor_id = r.floor_id
5 JOIN BED b ON r.room_id = b.room_id
6 WHERE b.bed_status = 'Available';
```

Code script

h_name	room_id	bed_id	bed_number
Hostel A	102	2	1
Hostel A	201	4	1
Hostel A	103	7	1
Hostel A	104	9	2
Hostel A	202	11	2
Hostel A	203	13	2
Hostel B	302	14	1
Hostel B	303	16	2

Output

Q2) Find the students who haven't made any payments?

Run SQL query/queries on table `student_hostel_management.student`: 

```
1 SELECT s.s_name
2 FROM STUDENT s
3 LEFT JOIN PAYMENT p ON s.student_id = p.student_id
4 WHERE p.student_id IS NULL;
```

Code script

Extra options

s_name

Sakib

Imran

Jahid

Rafi

Mehedi

Shuvo

Arman

Output

Q3) show students' name with their room and bed?

Run SQL query/queries on table student_hostel_management.STUDENT: ?

```
1 SELECT s.s_name, r.room_id, b.bed_number
2 FROM STUDENT s
3 JOIN ALLOCATION a ON s.student_id = a.student_id
4 JOIN BED b ON a.bed_id = b.bed_id
5 JOIN ROOM r ON b.room_id = r.room_id
6 WHERE a.end_date IS NULL;
```

Code script

s_name	room_id	bed_number
Riyad	101	1
Arafat	102	2
Siam	201	2
Nafi	301	1
Tanvir	104	1
Rakib	202	1
Hasan	203	1
Fahim	303	1

Output

Q4) show all payments made by Students

Run SQL query/queries on database student_hostel_management: ?

```
1 SELECT s.s_name, SUM(p.amount) AS total_payment
2 FROM STUDENT s
3 JOIN PAYMENT p ON s.student_id = p.student_id
4 GROUP BY s.s_name;
```

Code script

s_name	total_payment
Riyad	3000.00
Arafat	2500.00
Siam	2700.00
Nafi	2000.00
Tanvir	2500.00
Rakib	2700.00
Hasan	2000.00
Fahim	2100.00

Output

Q5) Find staff who handles most maintenance requests?

Run SQL query/queries on database student_hostel_management: ?

```
1 SELECT s.staff_name, COUNT(sm.request_id) AS total_requests
2 FROM STAFF s
3 JOIN STAFF_MAINTENANCE sm ON s.staff_id = sm.staff_id
4 GROUP BY s.staff_id
5 ORDER BY total_requests DESC
6 LIMIT 1;
```

Code script

staff_name	total_requests
Rahim	2

Output

Q6) Show all students with maintenance requests

Run SQL query/queries on database student_hostel_management: ?

```
1 SELECT s.s_name, m.request_id, m.description, m.m_status
2 FROM STUDENT s
3 JOIN ALLOCATION a ON s.student_id = a.student_id
4 JOIN BED b ON a.bed_id = b.bed_id
5 JOIN ROOM r ON b.room_id = r.room_id
6 JOIN MAINTENANCE_REQUEST m ON r.room_id = m.room_id;
```

Code script

s_name	request_id	description	m_status
Riyad	1	Fix fan	Pending
Arafat	2	Fix light	Completed
Siam	3	Fix water line	Pending
Nafi	4	Repair door	In Progress
Arafat	5	Fix fan noise	Completed

Output

Q7) Find the hostel with highest revenue

Run SQL query/queries on database student_hostel_management: ?

```
1 SELECT h.h_name, SUM(p.amount) AS total_revenue
2 FROM HOSTEL h
3 JOIN FLOOR f ON h.hostel_id = f.hostel_id
4 JOIN ROOM r ON f.floor_id = r.floor_id
5 JOIN BED b ON r.room_id = b.room_id
6 JOIN ALLOCATION a ON b.bed_id = a.bed_id
7 JOIN PAYMENT p ON a.student_id = p.student_id
8 GROUP BY h.h_name
9 ORDER BY total_revenue DESC
10 LIMIT 1;
```

h_name	total_revenue
Hostel A	15400.00

Q8) One student has done his hostel day student_id=4. Update the database allocation as inactive and set the end_date to today's date.

Run SQL query/queries on table student_hostel_management.student: ?

```
1 UPDATE ALLOCATION
2 SET status = 'Inactive',
3     end_date = CURDATE()
4 WHERE student_id = 4
5     AND status = 'Active';
```

Code script

✓ 1 row affected. (Query took 0.0059 seconds.)

```
UPDATE ALLOCATION SET status = 'Inactive', end_date = CURDATE() WHERE student_id = 4 AND status = 'Active';
```

[Edit inline] [Edit] [Create PHP code]

Output

Q9) Delete all complaint history from the database that were resolved and are older than 1 month.

Run SQL query/queries on table student_hostel_management.complaint: ?

```
1 DELETE FROM COMPLAINT
2 WHERE complaint_date < DATE_SUB(CURDATE(), INTERVAL 1 MONTH)
3 AND room_id IN (
4     SELECT room_id
5     FROM MAINTENANCE_REQUEST
6     WHERE m_status = 'Completed'
7 );
```

Code script

✓ 2 rows deleted. (Query took 0.0078 seconds.)

```
DELETE FROM COMPLAINT WHERE complaint_date < DATE_SUB(CURDATE(), INTERVAL 1 MONTH) AND room_id IN ( SELECT room_id FROM MAINTENANCE_REQUEST WHERE m_status = 'Completed' );
```

[Edit inline] [Edit] [Create PHP code]

Output

Business Process

Business process is a set of business activities put together with a goal of creating added value for a specific customer or market. Basic elements of a business process are the following: the goal, available resources, activities, indicators, focus on the buyer, and the process holders. There are some actors in this Student Hostel Management system :

- 1) Student
- 2) Hostel Admin
- 3) Staff Team
- 4) Finance Department

Some Function that adds extra value to the system are:

Student Registration Process: The admin registers students in the system. Student personal information is stored in the database. After checking their data accuracy, they become eligible for hostel.

Bed Allocation Process: Admin assigns a bed to one student based on availability of bed and this data is stored in ALLOCATION entity.

Student Payment Process: Students pay their hostel rent through different payment methods. For financial management purposes this data is stored in PAYMENT entity.

Complaint Management Process: Whenever any problem arises about service and facility , student can complain about those issues.

Maintenance Request processing: This service is for repairing any hostel components based on student requests.

This service supports all actors of this system on some criteria. A student can take one active bed at a time, and a bed would assign only one student at a time. Every payment is connected to one single student. Besides, complaints must be linked to student and room. Lastly, one staff member can handle multiple maintenance requests.